SEAT No.________

[36/A-16]

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SARDAR PATEL UNIVERSITY
BCA SEMESTER – IV EXAMINATION (OLD COURSE)
Monday, 7^{TH} September 2020
3.00 p.m. to 5.00 p.m.
US04FBCA01: COMPUTER BASED NUMERICAL AND STATISTICAL METHODS

Note: Scientific calculator is allowed. **Total Marks: - 70**

Q.1 Select appropriate answer from given choice. **10**

1. From the following _______ method is not iterative method.
   (a) False position (b) Bisection (c) Lagranges (d)none of them
2. If in a method of successive approximation, the root of equation lies between 1 and 2, $g(x) = \frac{1}{x^2-1}$, and initial guess is 1.25 then next approximation is
   (a) 0.5625 (b) 1.2177 (c) 1.7777 (d)none of them
3. Absolute Error is defined as __________
   (a) Present Approximation – Previous Approximation
   (b) True Value – Approximate Value
   (c) abs (True Value – Approximate Value)
   (d) abs (Present Approximation – Previous Approximation)
4. _______ is called the backward difference operator.
   (a) Δ (b) ∇(c) $\emptyset$ (d) $\cup$
5. To estimate the value of dependent variable x for a given value of independent variable y, the process is known as __________.
   (a) polynomial (b) Extrapolation (c) Interpolation (d) Inverse Interpolation
6. If the argument for the interpolation is not an equidistant then which method of interpolation is most appropriate.
   (a) Forward difference (b) backward difference
   (c) Newton's divided difference(d) all of the above
7. The system of linear equation $AX = B$ is said to be non-homogeneous if
   (a) $B \neq 0$(b) $B = 0$(c) $A \neq 0$(d) A is symmetric
8. For equally spaced tabular values, the Newton's forward interpolation formula can be used to find derivative at a point x if
   (a) x is in the upper half of the table (b) x is in the lower half of the table
   (c) x is at the center of the table (d) None of these
9. Irregular variations in a time series are caused by:
   (a) lockouts and strikes (b) epidemics
   (c) floods (d) all the above

10. Linear trend of a time series indicates towards:

(a) constant rate of change (b) constant rate of growth

(c) change in geometric progression (d) all the above

Q.2 True False? **8**

1. If the root of equation $x^3 - 2x - 5 = 0$ lies between (2,3) and if at i^{th} iteration $c = 2.5$ then next approximation by bisection method is given $c = \frac{2+2.5}{2}$.
2. False position method is the best method to obtain root ofnonlinearequation $f(x) = 0$.
3. Backward difference method is used if the estimated value lies towards the end of the difference table.
4. Moving average method is not a type of interpolation method.
5. We can find solution of system of linear, algebraic equations usingBisection method.

Fill in the blanks:

6. Rate of change of distance with respect to time represents_______.
7. Time series has ______ numbers of components.
8. Time series data are arranged according to _______.

Q.3 Attempt any Ten. **20**

1. Find the next approximation to the root of the non-linear equation$x^3 - 2x - 5 = 0$using false position method (Initial approximations are 1.75 and 2.5)
2. Explain following terms:

   (1) Relative error (2) Percentage Error
3. The solution of a problem is given as 3.436. It is known that the absolute error in the solution is less than 0.01. Find the interval within which the exact value must lie.
4. Write Newton's forward interpolation formula.
5. Define Interpolation.
6. Draw backward difference table.
7. If $x$lies in the lower half of the table and if $x = x_k$, then what is $\frac{dy}{dx}$ and$\frac{d^2y}{dx^2}$?
8. Write the system of equations in the matrix form:

   $2x - 2y + 5z = 13$, $2x + 3y + 4z = 20$, $3x - y + 3z = 10$
9. Solve the system of equations by using matrix inverse:

   $2x + 3y = 1$,

   $-x + 2y = 8$
10. What do you mean by Cyclic Variation?
11. List two utilities of time series.
12. List the components of Time series.

Q.4 Attempt any four. 32

(1) Using Bisection Method, find the approximate root of the non-linear equation $x^3 - 2x - 5 = 0$ Correct up to three decimal places.

(Take Initial approx. as 1.75 and 2.5)

(2) Using Newton Raphson method, find the approximate value of $\sqrt{13}$ correct up to 4 decimal places. (Take $x_0 = 3$)

(3) Given a function in the form of a table as

x	1	2	4
$y(x)$	1	5	9

Interpolate the value of $y(x)$ using Lagrangian polynomial at $x = 3$.

(4) From the following table, find y (2.5).

x	2	3	4	5
$y(x)$	5	9	15	25

(5) Solve the following system of equations by Gauss-Seidel method.

$$x + 2y + z = 0$$
$$3x + y - z = 0$$
$$x - y + 4z = 3$$

(6) Find $f'(0.85)$ from the following table:

x	0.50	0.75	1.00	1.25	1.50
$y = f(x)$	0.14	0.43	1.01	1.96	2.36

(7) Obtain the trend from the time series given below by 3- and 5- yearly moving average.

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
sales	100	95	91	83	70	65	63	55	50	46

(8) Obtain seasonal indices by simple average method.

Year	Q-1	Q-2	Q-3	Q-4
1	30	30	28	85
2	35	105	91	100
3	23	58	104	95
4	55	77	134	100

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(3)